

A Quick Look at Emerging Python Libraries for Data Scientists

If you are a data scientist, you need to stay abreast with everything that is happening in the field of data science. This article takes a brief look at Pandas Profiling and Streamlit – two emerging Python libraries that a data scientist should be aware of.



Data science is the buzzword these days, and Python is the language used most for it. Many new tools and libraries are emerging daily, which every data scientist should know about and use. In this article, we will take a look at two such libraries that will help a data scientist in exploratory data analysis (EDA) and model deployment.

Pandas Profiling

EDA is one of the critical steps in data science and gives users a certain level of insight and statistical measure of data. It also helps us to finalise the features for model development. Normally, we carry out EDA using Pandas and visualise data using Matplotlib, Seaborn or Bokeh, which is time consuming and requires a lot of programming skills. We can now do all of this by using the new library called Pandas Profiling.

Pandas Profiling is an open source Python library that helps to finish the standard EDA in a single line of code.

It also helps to generate profile reports from a Pandas data frame. It can be saved in Web format and shared within the team to validate and understand the data.

To install it using the pip package manager, use the following code:

```
pip install pandas-profiling
```

To generate the report, type:

```
profile = ProfileReport(df, title="Pandas Profiling Report")
```

The following code will generate the HTML report:

```
profile.to_file("report_name.html")
```

The report contains the following for each column in our data set: